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Electronic and vibrational structures in the S_0 and S_1 states of coronene

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We observed the fluorescence excitation spectra and dispersed fluorescence spectra of jet-cooled coronene- h_{12} and coronene- d_{12} . We analyzed the vibronic structures, assuming a planar and sixfold symmetric molecular structure (D_{6h}). The S_1 state was identified to be $^1B_{2u}$. The S_1 $^1B_{2u} \leftarrow S_0$ $^1A_{1g}$ transition is symmetry forbidden, so the 0_0^0 band is missing in the fluorescence excitation spectrum. We found a number of vibronic bands that were assigned to the e_{2g} fundamental bands and their combination bands with totally symmetric a_{1g} vibrations. This spectral feature is similar to that of benzene although several strong e_{2g} bands are seen in coronene. The band shape (rotational envelope) was significantly different in each e_{2g} mode. It was shown that degenerate rotational levels were shifted and split by the Coriolis interaction. We calculated the Coriolis parameter using the molecular structure in the S_1 state and the normal coordinate of each e_{2g} vibrational mode, which were obtained by theoretical calculations. The calculated band shapes well reproduced the observed ones, suggesting that the isolated coronene molecule has D_{6h} symmetry. *Published by AIP Publishing.*
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I. INTRODUCTION

Coronene (Fig. 1) is one of the most interesting polycyclic aromatic hydrocarbon molecules with high symmetry. Robinson and White reported the X-ray crystal structural analysis of coronene, which showed that the molecule is planar and has sixfold symmetry (D_{6h}).¹ The absorption spectrum was also well explained by assuming D_{6h} symmetry,² and the molecular structure was theoretically argued by Coulson.³ Ohno and Inokuchi, however, used fluorescence spectroscopy to show that the coronene molecule is distorted to have lower symmetry in the cold matrix.⁴ This symmetry lowering was also found in the excited triplet state by microwave-induced delayed phosphorescence.⁵

In order to investigate the structure of the isolated molecule, we performed careful observation of the fluorescence excitation and dispersed fluorescence spectra for coronene- h_{12} and coronene- d_{12} in supersonic jets. The observed spectra were much more accurate and had higher quality than the previously reported ones^{6,7} and the excitation spectrum inside He nanodroplets.⁸ The electronic and vibronic structures were elucidated by the detailed analysis of the results of the spectroscopic experiment and theoretical calculation.

We also investigated the Coriolis interaction between the degenerate rovibronic levels in the S_1 state. First we calculated the values of Coriolis interaction constants using the molecular structure and normal coordinates obtained by theoretical calculation. The rotational envelopes of vibronic bands were

then calculated by the obtained constants. The results properly reproduced the observed band shapes.

In this article, we present the results of laser spectroscopy for jet-cooled coronene- h_{12} and coronene- d_{12} and discuss the electronic and vibrational structures in the S_0 and S_1 states. Our experimental results are consistent with those of previous studies reporting that coronene has a planar molecular structure with sixfold symmetry.

II. EXPERIMENTAL

Commercially obtained coronene- h_{12} (Tokyo Chemical Industry, 95%) was repeatedly recrystallized from ethanol. Coronene- d_{12} (Cambridge Isotope Laboratory, 97%) was used without further purification. The solid sample was kept at 180 °C in a stainless steel container. The vapor was mixed with He gas (99.9%, 2 atm), and the mixed gas was expanded into a high-vacuum chamber through a pulsed nozzle (an automobile electric fuel injector) to generate a supersonic jet. The jet was crossed with a pulsed laser light beam at right angles. The distance between the nozzle and the crossing point was 1 cm.

As a light source, we employed a tunable dye laser (Lambda Physik, LPD3000, $\Delta E = 0.1 \text{ cm}^{-1}$, Exalite 416 and Exalite 392A) pumped by a pulsed excimer laser (Coherent, Compex Pro 110, $\Delta t = 5 \text{ ns}$, 308 nm, 200 mJ). The absolute wavenumber of laser light was calibrated using a wavemeter (Burleigh, WS-500) with the accuracy of $\pm 1 \text{ cm}^{-1}$.

Fluorescence from excited molecules was focused onto a photomultiplier surface (Hamamatsu R928) using a pair of lenses. The current output of the photomultiplier was amplified using a preamplifier (Stanford Research SR240). The electric

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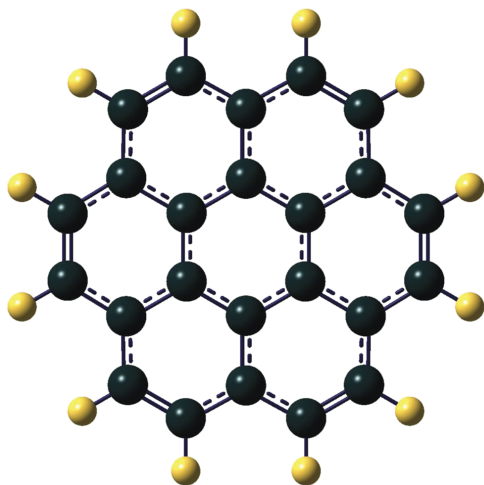


FIG. 1. Molecular structure of coronene.

signal of fluorescence intensity was obtained using a boxcar integrator (Stanford Research SR250), and a fluorescence excitation spectrum was obtained by recording the change in fluorescence intensity with the laser light wavelength. We could selectively observe the vibronic bands originating from coronene by picking out the signal of a long-lived component (200 μ s after the laser pulse) because the sample included a small amount of impurity that yielded vibronic bands in the same wavelength region. A dispersed fluorescence spectrum was observed using a scanning monochromator (Nikon P250) by tuning the laser light wavelength to a specific vibronic band.

III. RESULTS AND DISCUSSION

In this paper, the numbering of the fundamentals is uniformly chosen in accordance with the practice followed in the method shown in the work of Herzberg, unlike the conventional ways used in previous papers.^{4–7} The vibrations are grouped according to their irreducible representations, which

are taken in the order given by Wilson, Decius, and Cross.⁹ In the same symmetric group, vibrations are arranged in the order of their frequency, from high to low. The degenerated vibrations, however, are not counted doubly. Our numbering corresponds to Ohno's one⁴ as follows: ν_{34} to $\nu_1(e_{2g})$, ν_{31} to $\nu_3(e_{2g})$ and ν_{29} to $\nu_4(e_{2g})$, and ν_3 to $\nu_0(a_{1g})$.

The fluorescence excitation spectra of coronene- h_{12} and coronene- d_{12} in supersonic jets for 395–420 nm are shown in Fig. 2. A number of vibronic bands were also observed in the longer wavelength region, which were identified to originate from other compounds or their clusters because the fluorescence lifetimes were significantly short and the intensities were reduced by sample purification. The vibronic bands presented in Fig. 2 are therefore attributed to the $S_1 \leftarrow S_0$ transitions of coronene- h_{12} and coronene- d_{12} .

In order to assign each band, we performed theoretical calculations using the GAUSSIAN09 program package.¹⁰ First we calculated the stable molecular structure with geometry optimization assuming D_{6h} symmetry by DFT(CAM-B3LYP)/6-31G(d,p). The resultant structure was set as the initial input; we then performed the optimization again without any symmetry restriction. The result was approximately the same as the initial structure. The π orbitals are illustrated in Fig. 3. The wavefunctions are homogeneously distributed over the entire π frame except Φ_{73} and Φ_{84} where the π electrons are localized in the outer rim.

The lowest singlet excited state S_1 can be approximated with HOMO $\rightarrow$ LUMO single electron excitation. Both HOMO(e_{2u} ; Φ_{77} and Φ_{78}) and LUMO(e_{1g} ; Φ_{79} and Φ_{80}) are degenerate states, and the irreducible representations of the electronic excited states are obtained by the following direct product:

$$e_{2u} \otimes e_{1g} = B_{1u} + B_{2u} + E_{1u}. \quad (1)$$

This scheme is identical to that of benzene. The resultant three electronic states are actually split by configuration interaction, and the S_1 state is ${}^1B_{2u}$ as in benzene.

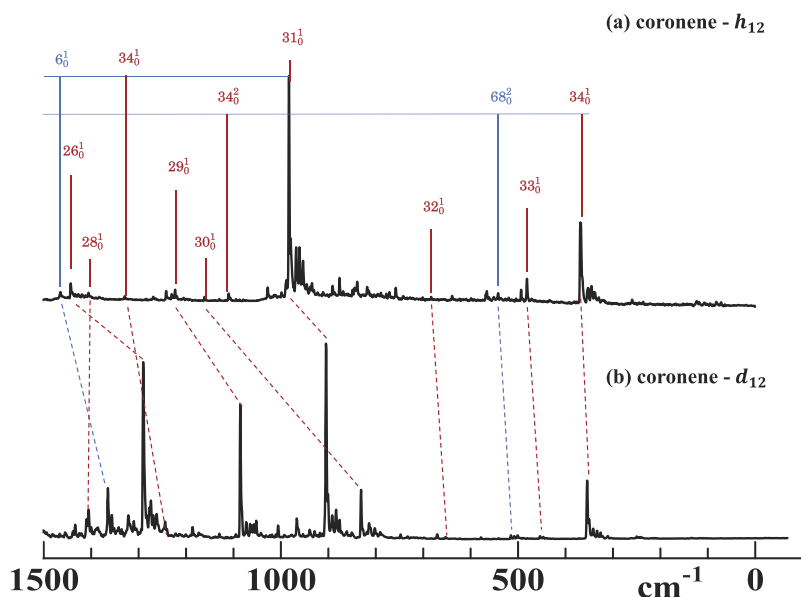


FIG. 2. Fluorescence excitation spectra of the S_1 ${}^1B_{2u} \leftarrow S_0$ ${}^1A_{1g}$ transition of jet-cooled (a) coronene- h_{12} and (b) coronene- d_{12} . The scale is the vibrational excess energy from the 0_0^0 transition.

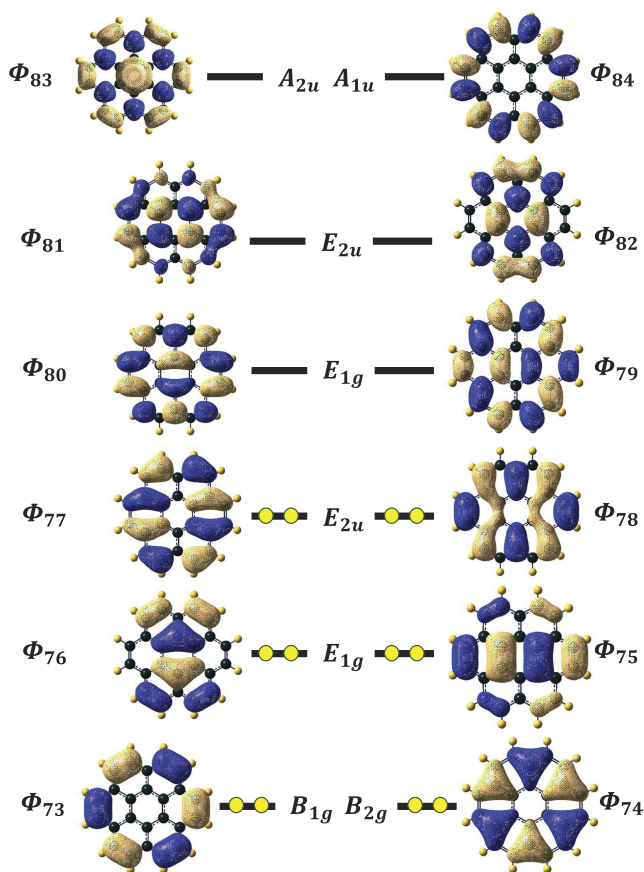


FIG. 3. Molecular orbitals of coronene.

For coronene, we calculated the excitation energies of the low excited states and the absorption oscillator strengths by SACCI(Full)/6-31G(d,p) using the optimized molecular structure obtained above. The results are tabulated in Table I. The S_1 state is ${}^1B_{2u}$ and the $S_1 {}^1B_{2u} \leftarrow S_0 {}^1A_{1g}$ transition is symmetry forbidden. It is expected from these results that the 0_0^0 band will be missing in the spectrum. The observed vibronic bands are considered to be observed by vibronic interaction with other excited states. Through Herzberg-Teller coupling scheme, the intensity is given by Refs. 11 and 12.

TABLE I. Excited states, excitation wavelengths, oscillator strengths, and coefficients of the main configurations of coronene calculated by SACCI(full)/6-31G(d,p).

Excited state	Excitation wavelength	Oscillator strength
$S_1 {}^1B_{2u}$	443.21 nm	$f = 0.0000$
	$\Phi_{77} \rightarrow \Phi_{80}$ -0.6432	
	$\Phi_{78} \rightarrow \Phi_{79}$ -0.6432	
$S_2 {}^1B_{1u}$	340.54 nm	$f = 0.0000$
	$\Phi_{77} \rightarrow \Phi_{79}$ 0.6656	
	$\Phi_{78} \rightarrow \Phi_{80}$ -0.6656	
$S_3 {}^1E_{1u}$	284.65 nm	$f = 1.1902$
	$\Phi_{77} \rightarrow \Phi_{79}$ -0.6386	
	$\Phi_{78} \rightarrow \Phi_{80}$ -0.6386	
	$\Phi_{77} \rightarrow \Phi_{80}$ 0.6386	
	$\Phi_{78} \rightarrow \Phi_{79}$ -0.6386	

$$I_{vr} \propto \left| \sum_k \sum_i \left(\langle \psi_e'' | \hat{\mu} | \psi_{ei} \rangle \left\langle \psi_{ei} \left| \frac{\partial}{\partial Q_k} \right| \psi_e' \right\rangle \right)_{Q_0} \langle \psi_v'' | Q_k | \psi_v' \rangle \right|^2, \quad (2)$$

where ψ_e'' , ψ_e' , and ψ_{ei} are the initial, final, and i th electronic wavefunction, ψ_v'' and ψ_v' are the initial and final vibrational functions, respectively. $\hat{\mu}$ is the electric dipole moment operator and Q_k are the k th normal coordinates. This is called “intensity borrowing,” in which a vibronic band of a forbidden transition shares the spectral intensity with another allowed transition. This is possible for the vibronic level of the S_1 state and other excited states with the same symmetry (irreducible representation). From the S_0 state of the D_{6h} molecule, the $\pi\pi^*$ electronic transition is allowed only to the ${}^1E_{1u}$ state. Consequently, the e_{2g} vibrational levels in the $S_1 {}^1B_{2u}$ state become vibronically active in the electronic transition from the zero-vibrational level in the S_0 state (Fig. 4).

In order to assign the observed vibronic bands, we calculated the vibrational energies in the S_1 state of coronene- h_{12} and coronene- d_{12} by TDDFT(CAM-B3LYP)/6-31G(d,p). The results for the a_{1g} and e_{2g} modes are listed in Table II. The observed vibronic band of the smallest wavenumber is assigned to ν_{34} (skeletal deformation), which is the least energetic e_{2g} mode. We then anticipated the position of the 0_0^0 band using the calculated value of the vibrational energy. Next, we assigned other observed bands. The calculated vibrational energies were appreciably larger than the observed values, however. This discrepancy generally occurs because anharmonicity is not incorporated in the theoretical calculation. Here, we introduced a scale factor to properly reduce the calculated vibrational energy. This procedure was iteratively repeated and we eventually obtained the best-fit 0_0^0 position and scale factor. The transition wavenumber of the 0_0^0 bands is expected to be 23 802 and 23 874 cm^{-1} for coronene- h_{12} and coronene- d_{12} , respectively. The 0_0^0 band is generally blue-shifted by deuterium substitution because the reduction of zero-point energy in the S_1 state is smaller than that in the S_0 state.¹³ The assignments of the observed bands are listed in

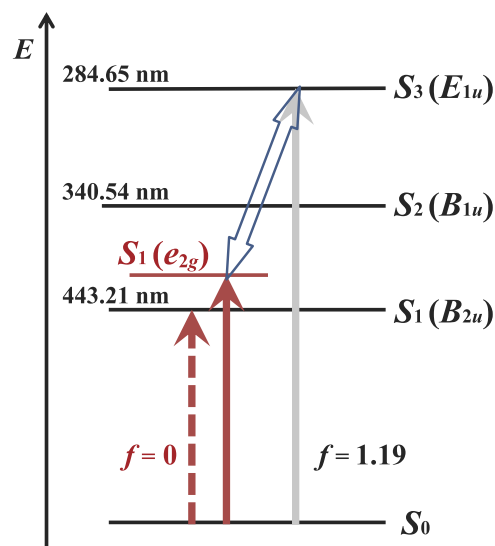


FIG. 4. Scheme of the vibronic transition of coronene.

TABLE II. Calculated main vibrational energies (cm^{-1}) of coronene- h_{12} and coronene- d_{12} .

Number	Symmetry	Calc. (obs. ^a)			
		Coronene- h_{12}		Coronene- d_{12}	
		S_0^b	S_1^c	S_0^d	S_1^c
1	a_{1g}	3125	3113	2294	2304
2	a_{1g}	1644	1583	1589	1541
3	a_{1g}	1375 (1355)	1367 (1379)	1357 (1356)	1360
4	a_{1g}	1225	1214	815	815
5	a_{1g}	1037	1033	1092	1106
6	a_{1g}	480 (488)	478 (482)	460 (459)	462 (460)
23	e_{2g}	3122	3110	2292	2302
24	e_{2g}	3106	3094	2268	2281
25	e_{2g}	1665	1658 (1637)	1634	1646
26	e_{2g}	1473 (1458)	1447 (1441)	1305 (1302)	1306 (1288)
27	e_{2g}	1453	1489	1414	1474
28	e_{2g}	1413 (1421)	1396 (1404)	1433 (1422)	1415 (1403)
29	e_{2g}	1229	1242 (1221)	1089 (1096)	1089 (1083)
30	e_{2g}	1161	1146 (1159)	823 (830)	825 (828)
31	e_{2g}	997 (996)	975 (981)	902 (915)	899 (902)
32	e_{2g}	677	673 (682)	645	648 (650)
33	e_{2g}	487	483 (483)	444	445 (451)
34	e_{2g}	363 (371)	359 (367)	352 (359)	352 (352)
68	e_{2u}	86	86 (87)	80 (90)	81 (81)

^aDetermined using the fluorescence excitation spectrum with an accuracy of $\pm 1 \text{ cm}^{-1}$ in S_1 and using the dispersed fluorescence spectrum with an accuracy of $\pm 12 \text{ cm}^{-1}$ in S_0 .

^bResults of DFT(CAM-B3LYP)/6-31G(d,p) are scaled by 0.970 in S_0 of coronene- h_{12} .

^cResults of TDDFT(CAM-B3LYP)/6-31G(d,p) are scaled by 0.965 in S_1 of coronene- h_{12} and coronene- d_{12} .

^dResults of DFT(CAM-B3LYP)/6-31G(d,p) are scaled by 0.960 in S_0 of coronene- d_{12} .

Table III, and indicated in Fig. 2. The calculated vibrational energies with the best-fit scale factor are in good agreement with the observed values. In the fluorescence excitation spectrum of coronene- d_{12} , five bands are prominent and all are assigned to the e_{2g} fundamental bands. These are also identified in the spectrum of coronene- h_{12} although some bands are markedly weak. Other weak bands are assigned to the combination bands with the a_{1g} mode.

We observed the dispersed fluorescence spectrum by tuning the laser wavelength to an e_{2g} vibronic band. The results of coronene- d_{12} are shown in Fig. 5. The observed vibronic bands are assigned in the same manner as the fluorescence excitation spectrum. The vibrational energies in the S_0 state were calculated by DFT(CAM-B3LYP)/6-31G(d,p) with geometry optimization. The results for the a_{1g} and e_{2g} modes are listed in Table II. Using these values, we could assign the observed

TABLE III. Observed and calculated vibrational energies (cm^{-1}) and assignments in the S_1 $^1B_{2u}$ states of coronene- h_{12} and coronene- d_{12} .

Coronene- h_{12} obs. ^a (calc. ^b)	τ_f (ns)	Coronene- d_{12} obs. ^a (calc. ^b)	τ_f (ns)	Assignment
0		0		0_0^0 (h_{12} : 23 802 cm^{-1} , d_{12} : 23 874 cm^{-1}) ^c
367 (359)	452	352 (352)	580	$34_0^1(e_{2g})$
479 (483)		451 (445)		$33_0^1(e_{2g})$
540 (524)		513 (513)		$34_0^1 68_0^2$
682 (673)		650 (648)		$32_0^1(e_{2g})$
981 (975)	382	902 (899)	620	$31_0^1(e_{2g})$
1108 (1077)				34_0^3
1159 (1146)		828 (825)	610	$30_0^1(e_{2g})$
1221 (1242)		1083 (1089)	550	$29_0^1(e_{2g})$
1404 (1396)		1403 (1415)		$28_0^1(e_{2g})$
1441 (1447)		1288 (1306)	520	$26_0^1(e_{2g})$
1463 (1453)		1362 (1361)		$31_0^1 6_0^1$
		1452 (1467)		$26_0^1 68_0^2$
		1543 (1551)		$29_0^1 6_0^1$

^aDetermined using the fluorescence excitation spectrum with an accuracy of $\pm 1 \text{ cm}^{-1}$.

^bResults of TDDFT(CAM-B3LYP)/6-31G(d,p) are scaled by 0.965.

^cEstimated by a least-squares fit of the observed vibrational energies to the calculated ones. See text.

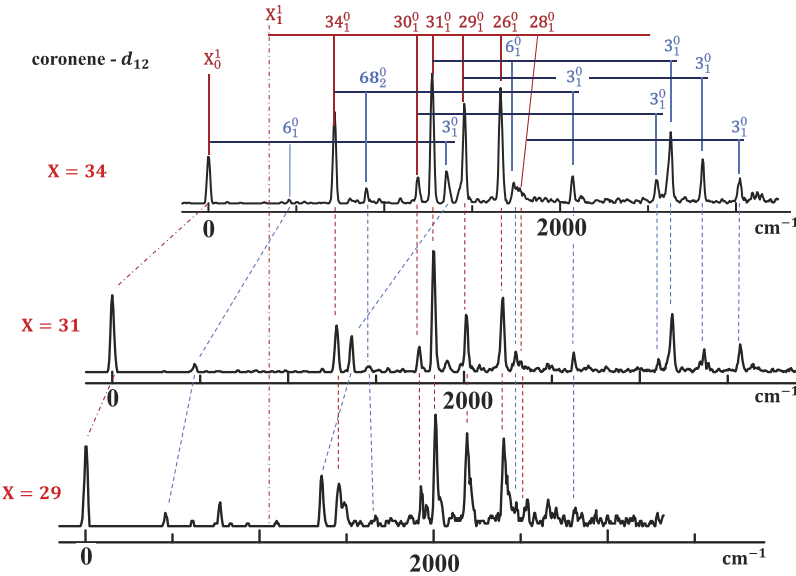


FIG. 5. Dispersed fluorescence spectra of the coronene- d_{12} . X is an excited vibration, and the scale is the vibrational energy from the X_0^1 band.

vibronic bands, and the final assignments are shown in Table IV and Fig. 5. It is noted that the spectrum for the high-energy level excitation shows a broad component and the same spectral pattern as that for the low-energy level excitation. This is evidence of intramolecular vibrational redistribution (IVR), which occurs at the high-vibrational levels in the S_1 state.^{13–15} The coronene molecule is markedly large and the vibrational level density is high. Consequently, IVR is considered to

efficiently occur, even at the vibrational energy of approximately 1000 cm^{-1} .

The normal coordinates of active modes in the spectra are illustrated in Fig. 6. The e_{2g} modes are mostly deformation vibrations over the entire molecular frame. The 34_0^1 and 31_0^1 bands are prominent in the fluorescence excitation spectrum of coronene- h_{12} . In coronene- d_{12} , the number of strong bands increases and the 34_0^1 , 31_0^1 , 30_0^1 , 29_0^1 , and 26_0^1 bands are

TABLE IV. Observed and calculated vibrational energies (cm^{-1}) and assignments in the S_0 $^1A_{1g}$ states of coronene- h_{12} and coronene- d_{12} .

Coronene- h_{12}		Obs. ^a (calc. ^b)			Assignment ^c
X = 34	X = 31	X = 34	X = 31	X = 29	
488 (480)		456 (461)	468 (461)	458 (460)	$X_0^1 6_1^0$
1367 (1375)	1361 (1360)	1352 (1357)	1363 (1357)	1357 (1357)	$X_0^1 3_1^0$
	(371)	(359)	(913)	(1096)	(X_1^1)
	1361 (1360)		1275 (1255)	1456 (1442)	$X_1^1 34_1^0$
		1189 (1176)	1746 (1726)	1927 (1912)	$X_1^1 30_1^0$
1367 (1360)		1273 (1256)		2011 (1991)	$X_1^1 31_1^0$
		1454 (1442)	2011 (1992)		$X_1^1 29_1^0$
1792 (1775)	2416 (2409)	1789 (1787)	2317 (2337)	2536 (2522)	$X_1^1 28_1^0$
1829 (1835)	2449 (2469)	1661 (1659)	2223 (2208)	2404 (2394)	$X_1^1 26_1^0$
		897 (871)	1455 (1421)	1666 (1607)	$X_1^1 34_1^0 68_2^0$
1829 (1840)		1737 (1716)		2475 (2452)	$X_1^1 31_1^0 6_1^0$
		2119 (2119)			$X_1^1 26_1^0 6_1^0$
			2472 (2453)		$X_1^1 29_1^0 6_1^0$
			2625 (2613)	2815 (2798)	$X_1^1 34_1^0 3_1^0$
		2545 (2534)	3105 (3083)		$X_1^1 30_1^0 3_1^0$
2726 (2734)		2073 (2063)			$X_1^1 31_1^0 3_1^0$
		2810 (2800)	3366 (3350)		$X_1^1 29_1^0 3_1^0$
			3645 (3694)		$X_1^1 28_1^0 3_1^0$
3173 (3209)	3842 (3843)	3023 (3016)	3570 (3566)		$X_1^1 26_1^0 3_1^0$
742 (726)	2006 (1994)	718 (705)	1825 (1805)	2192 (2178)	X_2^1
	2496 (2474)		2294 (2265)	2658 (2638)	$X_2^1 6_1^0$
2093 (2100)	3363 (3368)	2073 (2063)	3184 (3163)		$X_2^1 3_1^0$

^aDetermined using the fluorescence excitation spectrum with an accuracy of $\pm 12\text{ cm}^{-1}$.

^bResults of DFT(CAM-B3LYP)/6-31G(d,p) are scaled by 0.970 for coronene- h_{12} and 0.960 for coronene- d_{12} .

^c X is an excited vibration.

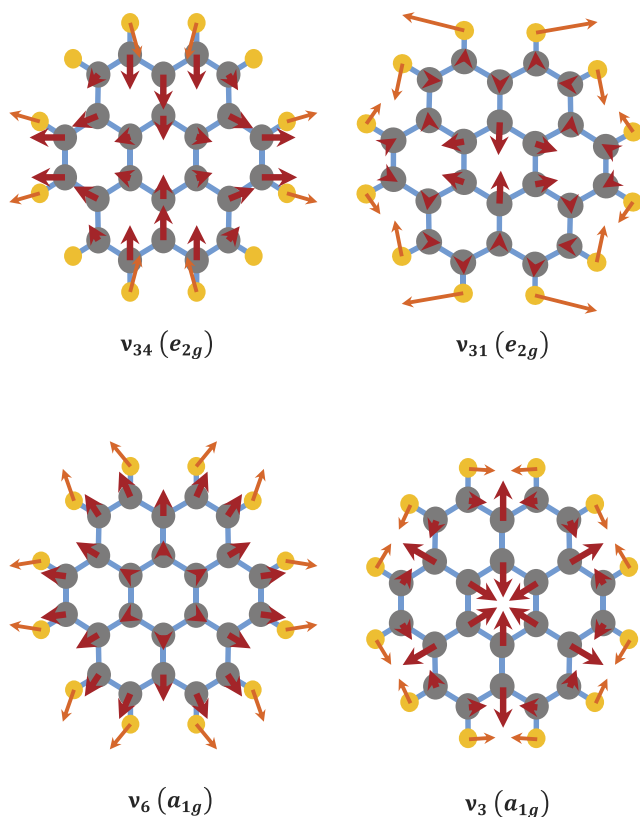


FIG. 6. Normal coordinates of the main vibrations of coronene.

comparatively strong. This indicates that the nonadiabatic Franck-Condon factor becomes larger by deuterium substitution for more e_{2g} modes. In ν_{34} and ν_{31} , the vibrational motion on the pivot ring is similar to that of benzene ν_6 , suggesting that the displacements of the C atoms on the pivot ring are important for the vibronic band intensity. Two a_{1g} modes, ν_6 , and ν_3 are also important for the intensity of combination bands. The normal coordinate of ν_6 can be called “breathing” like benzene ν_1 , which is considered to be the in-phase expansion of the entire molecule. In ν_3 , however, the vibrational motion of the outer-rim C atoms is out-of-phase and the outer phenyl rings move like they do in Kekule vibration.

Here, we consider the nonradiative transition and fluorescence quantum yield in the S_1 $^1B_{2u}$ state of coronene. The fluorescence quantum yield Φ_f is given by Ref. 16,

$$\Phi_f = \frac{k_r}{k_r + k_{nr}} = \frac{1/\tau_r}{1/\tau_r + 1/\tau_{nr}} = \frac{1/\tau_r}{1/\tau_f}, \quad (3)$$

where k_r and k_{nr} are the radiative and nonradiative decay rate, respectively, and τ represents each lifetime. τ_f is the fluorescence lifetime, which can be obtained by the observed fluorescence decay curve. If the nonradiative transition is fast, τ_f is short and Φ_f is low. Although it is very difficult to experimentally determine the τ_r value, it has been estimated to be 1 μ s in benzene by extensive studies.^{17,18} The fluorescence lifetime was accurately measured to be approximately 100 ns for a number of low-energy vibrational levels in benzene. If we assume that τ_r in the S_1 state of coronene is also 1 μ s, the Φ_f values are estimated to be 0.45 and

0.58 for coronene- h_{12} and coronene- d_{12} , respectively, by the observed fluorescence lifetime shown in Table III. The nonradiative lifetime generally becomes longer by deuterium substitution. This suppression of nonradiative transition is attributed to the small overlap of wavefunctions between the coupling levels.

It is expected that the nonradiative decay becomes faster as the molecule becomes larger because the vibrational level density becomes larger.¹⁹ The nonradiative decay is actually, however, not very fast in coronene—this is the same in large pericondensed aromatic hydrocarbons such as pyrene¹³ and perylene.²⁰ It is attributed to the small change in the molecular structure upon the $S_1 \leftarrow S_0$ electronic transition. The change in the π bond order is considered to be diluted with a large number of π electrons in large aromatic hydrocarbons. If the two potential energy curves of the S_0 and S_1 states are identical, the nonadiabatic Franck-Condon factor is zero and internal conversion (IC) never occurs.^{21,22} Intersystem crossing (ISC) to the triplet is also assumed to become faster with increased molecular size. The ISC rate is, however, originally very small in the isolated molecule of a planar aromatic hydrocarbons (El-Sayed rule).^{23,24} The drastic enhancement of nonradiative decay in the high vibrational level of the S_1 state, which is clearly seen in benzene (channel three), is of note.^{25,26} We could not confirm this in coronene up to the vibrational energy of 1500 cm^{-1} . Accurate measurements of fluorescence lifetime in the higher energy region are needed.

Next, we discuss the Coriolis interaction between the degenerate e_{2g} vibrational levels. Expanded fluorescence excitation spectra for the e_{2g} fundamental bands are shown in Fig. 7. The band shape is an envelope of overlapped rotational lines. The observed band shape was markedly different from the vibronic band, indicating that the Coriolis interaction differs in each e_{2g} mode. If the Coriolis interaction is very small, the band shape is almost identical for all vibronic bands, with a couple of broad peaks of P and R branches and a dip at the band center. This is a typical shape of a perpendicular band with an oblate symmetric top. The principal rotational axis (c) is out-of-plane, whereas the $\pi\pi^*$ transition moment is in-plane. The Coriolis interaction results from the coupling between a vibrational angular momentum $\mathbf{P}$, which arises from the degenerate vibrational motions in different phases, and a rotational angular momentum $\mathbf{J}$. The perturbative Hamiltonian is expressed by Refs. 27 and 28,

$$H_{\text{Cor}} = -\frac{P_x J_x}{I_x} - \frac{P_y J_y}{I_y} - \frac{P_z J_z}{I_z}, \quad (4)$$

where P_i is the component of vibrational angular momentum and J_i is that of rotational angular momentum. I_i is the component of the moment of inertia. The Coriolis interaction between the E_{2g} degenerate levels (r and s) occurs around the out-of-plane c axis (We take this as the z axis.). Therefore,

$$H_{\text{Cor}} = -\frac{P_z J_z}{I_z} = -2C\zeta_{r,s}^z(Q_r P_s - P_r Q_s), \quad (5)$$

where C is the rotational constant, and Q_r and Q_s are the normal coordinates of two degenerated vibrations. $\zeta_{r,s}^z$ is called the Coriolis zeta constant, which represents the coupling strength of the Coriolis interaction and is given by

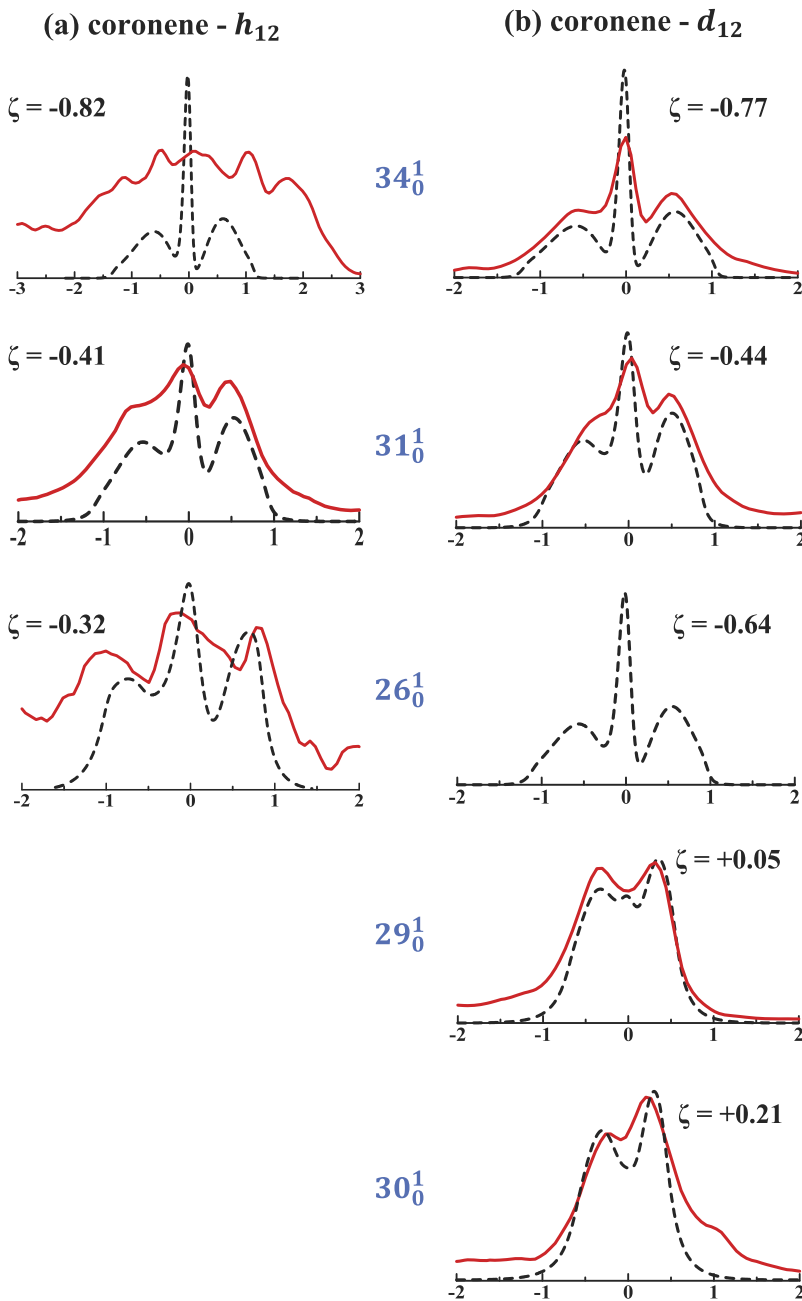


FIG. 7. Observed (red) and calculated (black) rotational contours of vibronic bands with calculated Coriolis constants. See text for the details of the calculation.

$$\zeta_{r,s}^z = - \sum_n \left[\left(\frac{\partial x_n}{\partial Q_r} \right) \left(\frac{\partial y_n}{\partial Q_s} \right) - \left(\frac{\partial x_n}{\partial Q_s} \right) \left(\frac{\partial y_n}{\partial Q_r} \right) \right]. \quad (6)$$

x_n and y_n are the mass-adjusted Cartesian displacements of the n th atom, which are obtained by the Cartesian displacements X_n and Y_n and the atomic mass m_n ,

$$x_n = \sqrt{m_n} X_n, \quad y_n = \sqrt{m_n} Y_n. \quad (7)$$

The vibrational angular momentum is obtained by the operator

$$P_z = - \sum_n i\hbar \left[x_n \left(\frac{\partial}{\partial y_n} \right) - y_n \left(\frac{\partial}{\partial x_n} \right) \right]. \quad (8)$$

The rotational levels are shifted by this interaction. The following experimental Coriolis parameter is used to calculate the actual level energies:

$$\xi_{r,s}^c = |2C\zeta_{r,s}^z|. \quad (9)$$

We designate the vibrational and rotational term values (in units of cm^{-1}) as $G(r) = G(s)$ and $F(J, K_a, K_c)$, respectively. The energy matrix is given by

$$\begin{pmatrix} G(r) + F(J, K_a, K_c) & i\xi_{r,s}^c K_c \\ -i\xi_{r,s}^c K_c & G(s) + F(J, K_a, K_c) \end{pmatrix}. \quad (10)$$

The eigenvalues of two rovibronic levels are obtained by diagonalizing this.

We first calculated the Coriolis zeta constant using the results of TDDFT(CAM-B3LYP)/6-31G(d,p) with geometry optimization. We obtained the matrix elements from the optimized molecular structure and the normal coordinates. We assumed the D_{6h} symmetry in this DFT calculation, and therefore $A = B = 2C$. The rotational envelope for each e_{2g} fundamental band was calculated using the PGOPHER program.²⁹ The results were compared with the observed band

shape as shown in Fig. 7. The calculated band shapes approximately reproduced the observed ones. The shape of the E_{1u} vibronic band in coronene is well explained by the Coriolis interaction, and the strength can be accurately estimated by the molecular structure and normal coordinates obtained by theoretical calculation, except the 34_0^1 band of coronene- h_{12} which seems to be splitting into two bands. This cause is unclear.

We consider that the coronene molecule shows D_{6h} symmetry because we could find no evidence of large structural distortion. If the molecule were bent out-of-plane, the 0_0^0 band would be observed to have a sharp Q -branch peak at the center (parallel transition).³⁰ If the molecule were distorted in-plane, the degenerate vibrational levels would be split. We did not observe any such anomalous spectral features.

V. SUMMARY

We observed the fluorescence excitation and dispersed fluorescence spectra of coronene- h_{12} and coronene- d_{12} in supersonic jets using a nanosecond pulsed dye laser. A number of bands were observed around 410 nm, which were identified to be the vibronic bands of the $S_1^1B_{2u} \leftarrow S_0^1A_{1g}$ transition. Prominent bands are the e_{2g} fundamental bands and their combination bands with the a_{1g} modes. The intensity arises from the vibronic coupling with the $S_3^1E_{1u}$ state. It was found that the shape of e_{2g} fundamental bands was markedly different in the vibrational mode. The observed band shape is well reproduced by the simulation with the calculated Coriolis zeta constant. These experimental results indicate that the coronene molecule shows approximately D_{6h} symmetry. Rotationally resolved ultrahigh-resolution spectroscopy is needed to estimate the molecular structure and elucidate the excited-state dynamics.

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